

Van-Loc (Felix) Nguyen

Ho Chi Minh City, Vietnam

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EXPERIENCE

- **Eurofins** Ho Chi Minh City, Vietnam
Software Engineer Apr 2026 – Present
 - **IT application monitoring and observability.**
- **National Institute of Informatics** Chiyoda-ku, Tokyo, Japan
Research Intern Oct 2025 – Mar 2026
 - Investigated **synthetic-image detection using compact vision-language models** at Professor Isao Echizen's Laboratory as part of the NII International Internship Program.
 - Designed and conducted **synthetic-image classification experiments** with **SmolVLM, Qwen, and InternVL** to evaluate their ability to distinguish AI-generated images from real images.
 - Analyzed model errors and experimental results, identifying limitations of compact vision-language models for reliable synthetic-image detection.
 - **Mentor:** Dr. Pyone MaungMaung
- **Autonomous Inc** Ho Chi Minh City, Vietnam
AI Engineer Mar 2025 – Aug 2025
 - Developed **LLM-powered agents** for production applications using **OpenAI APIs and LangChain**, including tool calling and multi-step workflows.
 - Improved the reliability of **locally deployed agents on macOS** by designing context-management and prompt-engineering strategies for compact models, including Qwen 4B.
 - Optimized agent workflows for **resource-constrained environments** by reducing unnecessary context and improving tool-selection consistency.
- **Saritasa** Ho Chi Minh City, Vietnam
Backend Engineer Jun 2023 – Feb 2025
 - Developed and maintained **Django REST APIs** and asynchronous **Celery workflows** for several US-market products using Python and PostgreSQL, including a production application that reached **1,000 users within two months of launch**.
 - Improved backend performance and reliability by optimizing database queries, API endpoints, and background-processing workflows.
 - Diagnosed and resolved production issues across **Kubernetes-hosted services**, collaborating with distributed engineering and product teams throughout the delivery lifecycle.
- *Backend Engineer Intern* Mar 2023 – May 2023
 - Built and tested backend features with **Python, Django, PostgreSQL, and Docker** while completing Saritasa's backend engineering training program.

EDUCATION

- **University of Science, VNUHCM** Ho Chi Minh City, Vietnam
Master of Science in Computer Science, Artificial Intelligence; GPA: 4.00/4.00 Dec 2024 – Present
 - **Relevant Coursework:** Mathematics for AI, Machine Learning, and Natural Language Processing.
- **University of Science, VNUHCM** Ho Chi Minh City, Vietnam
Bachelor of Science in Computer Science, Honors Program; GPA: 3.88/4.00 Oct 2020 – Oct 2024
 - **Graduation Thesis:** Fake News Analysis and Detection; Thesis Score: 10.00/10.00 (4.00/4.00).

SELECTED PROJECTS

- **Doodle Duel: AI-Powered Multiplayer Drawing Game** Feb 2026
Main Developer *Deployment* — *GitHub*
 - Built a real-time multiplayer drawing game at Gemini 3 Hackathon Tokyo using React, Express, TypeScript, and Socket.IO with live canvas synchronization, a round timer, and scoring.
 - Integrated **Gemini 2.5 Flash** vision for sketch classification and LLM-based refereeing to determine winners; used **Imagen 4** for prompt and image generation.
 - Deployed a containerized full-stack system with Docker, Traefik, and Let's Encrypt.
- **ArcanaAI: AI-Powered Tarot Reading Service** Apr 2025 – Aug 2025
Main Developer *Deployment* — *GitHub*
 - Built an AI-powered tarot reading platform with GPT-based real-time interpretations using OpenAI function calling.
 - Developed interactive chat, journaling, and history features with FastAPI, LangChain, Next.js, and Docker.
- **GraphRAG: Knowledge-Graph Retrieval System** Mar 2025
Main Developer *GitHub*
 - Built a RAG system combining vector search and knowledge-graph traversal with local LLMs.
 - Automated triplet extraction and graph construction from documents.
 - Developed a hybrid vector and knowledge-graph retrieval strategy to improve accuracy and reduce hallucination.
 - Deployed the system with FastAPI and a minimal web interface.
- **Fake News Analysis and Detection** Feb 2024 – Aug 2024
Co-Main Developer; Undergraduate Thesis *GitHub*
 - Completed at the University of Science, VNUHCM.
 - Proposed an end-to-end network for the Visual Entailment task to solve both tasks of the Cheapfakes Detection Challenge.
 - Introduced a cheapfakes data-augmentation method using LLMs.
 - Implemented a reputation-checking module that searches for images on the Internet to utilize social information.
 - Developed an open system that helps users analyze and detect fake news with an external forensic framework.

SKILLS

- **Programming Languages:** Python, C++, TypeScript, SQL.
Backend: FastAPI, Django, Django REST Framework, Celery, PostgreSQL, Express, Socket.IO.
AI/ML: PyTorch, LangChain, OpenAI API, vision-language models, RAG.
Infrastructure: Docker, Kubernetes, Traefik, Linux.
Spoken Languages: Vietnamese, English.

PUBLICATIONS

- **Google Scholar:** scholar.google.com/citations?user=39z1A1IAAAAJ

CERTIFICATES & AWARDS

- **Third Prize, The 26th Student Scientific Research Award – Euréka 2024:** Ho Chi Minh City, Vietnam
- **Third Prize, The 2024 “AI City Challenge” (AIC) of Ho Chi Minh City:** Ho Chi Minh City, Vietnam
- **Certificate of Satisfactory Progress from President of VNUHCM:** Ho Chi Minh City, Vietnam